# The Tree and the Shadow:

## Geometric Derivation of the Reciprocal Silver Ratio $\sqrt{2}-1$ and an Exploratory Statistical Analysis of Tree Crown Ratios

A Pilot Study Bridging Geometry, Biology, and Number Theory

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**Abstract**

This paper presents the Al-Hindara framework as a conceptual approach to connecting geometric modeling, management, and empirical observation. It derives the exact trigonometric identity

$$\tan\left(\frac{\pi}{8}\right) = \sqrt{2}-1,$$

the reciprocal of the conventional Silver Ratio $1+\sqrt{2}$. The paper then explores whether this geometric reference value is consistent with the crown-to-total-height ratio of five mango trees reported to have been measured in Sho'ub, Sana'a, Yemen.

The reported pilot sample has a mean crown ratio of approximately 0.41297. A Two One-Sided Tests (TOST) procedure is described using an equivalence margin of $\pm 0.03$ around the theoretical reference value. Under the assumptions of the test, the reported results indicate that the sample mean falls within this prespecified equivalence region. The small sample size and single-location sampling limit the interpretation: these findings do not establish a universal biological law or a causal relationship between solar geometry and tree crown structure.

The paper also introduces a conjectural numerical modeling proposal involving the integer 21, obtained by taking the floor of the ordinate of the second positive nontrivial zero of the Riemann zeta function. The proposed numerical connection is exploratory and is not presented as a theorem about the zeta function.

The paper distinguishes an exact geometric identity from empirical observations and conjectural numerical constructions. Its principal aim is to provide a transparent basis for further mathematical and field investigation.

**Keywords:** Reciprocal Silver Ratio, Half-Angle Identity, Crown Ratio, TOST Equivalence Testing, Mango Trees, Riemann Zeta Function, Al-Hindara.

## 1 Introduction

The Al-Hindara framework introduces the conceptual relation

$$Z + C + A = 1, \tag{1}$$

where $Z$, $C$, and $A$ are intended to represent normalized components associated with geometry, management, and synthesis.

In this paper, Equation (1) is treated as a conceptual normalization proposal. The variables must be operationally defined and normalized before the equation can be tested as a quantitative empirical relationship. It is not a consequence of the Pythagorean theorem, and the present study does not establish it as a universal law.

The paper has three objectives. First, it derives the exact trigonometric value associated with a 22.5° angle measured from the vertical. Second, it compares that value with the reported crown-to-total-height ratios of five mango trees. Third, it outlines a conjectural numerical construction involving an integer derived from a nontrivial zero ordinate of the Riemann zeta function.

These objectives have different evidential status. The trigonometric identity is an exact mathematical result. The tree measurements constitute a small exploratory dataset. The proposed number-theoretic connection is a conjecture that requires a separate mathematical derivation.

## 2 Geometric Derivation of the Reciprocal Silver Ratio

Consider an idealized right triangle in which $Z$ denotes the vertical height, $C$ the horizontal shadow length, and $A$ the length of the slanted ray. The Pythagorean theorem gives

$$Z^2 + C^2 = A^2. \tag{2}$$

Let $\theta$ be the angle between the vertical direction and the slanted ray. Under this convention, the side opposite $\theta$ is $C$, and the adjacent side is $Z$. Therefore,

$$\tan(\theta) = \frac{C}{Z}. \tag{3}$$

This angle convention differs from the conventional solar elevation angle measured from the horizontal. If $\theta = 22.5°$ from the vertical, the corresponding elevation angle from the horizontal is 67.5°.

To evaluate the ratio at $\theta = 22.5°$, use the half-angle identity

$$\tan\left(\frac{\phi}{2}\right) = \frac{1 - \cos\phi}{\sin\phi}. \tag{4}$$

Setting $\phi = 45°$ yields

$$\tan(22.5°) = \frac{1 - \frac{\sqrt{2}}{2}}{\frac{\sqrt{2}}{2}} \tag{5}$$

$$= \frac{2 - \sqrt{2}}{\sqrt{2}} \tag{6}$$

$$= \sqrt{2} - 1 \tag{7}$$

$$\approx 0.4142135624. \tag{8}$$

The conventional Silver Ratio is $1 + \sqrt{2}$. Since

$$\frac{1}{1 + \sqrt{2}} = \sqrt{2} - 1,$$

the value in Equation (8) is its reciprocal.

This result is exact for the specified idealized geometry. It does not imply that tree crown structure must follow the same ratio. Any application to biological morphology is an empirical hypothesis that requires independent observation.

# 3 Exploratory Field Measurement Protocol

## 3.1 Study objective and operational definition

The exploratory analysis compares the geometric reference value with a crown ratio defined by

$$R = \frac{H - h}{H}, \tag{9}$$

where $H$ is the total tree height and $h$ is the height of the first main branching point, both measured from the same ground-level reference.

The variable $R$ is dimensionless. It represents the fraction of total height above the specified branching point under this operational definition. It should not automatically be interpreted as a complete measure of crown volume, crown width, or foliage density.

The target reference is

$$R_0 = \sqrt{2} - 1 \approx 0.4142135624. \tag{10}$$

## 3.2 Reported study setting and measurements

The supplied study record describes a pilot sample of five mango trees (*Mangifera indica*) in Sho'ub, Sana'a, Yemen, on October 14, 2026.

The reported measurement procedure specifies a metal measuring tape with a stated precision of $\pm 1$ mm and an extendable rigid pole used to align the measurement with the highest leaf apex. The first main branching point is defined as the first clear bifurcation at which each of the two branches reaches at least 30% of the original trunk diameter.

These procedural details should be retained as a description of the actual study only if they accurately reflect the measurements performed. The nominal precision of the tape does not, by itself, establish the total accuracy of a 7–9 m tree-height measurement. Alignment, access to the highest leaf, ground slope, branch identification, and repeatability may contribute additional uncertainty.

For reproducibility, a future study should document the measurement instrument, the positioning procedure, the number of repeat measurements, the treatment of sloping ground, and the uncertainty associated with both $H$ and $h$.

## 3.3 Raw pilot data

Table 1 reproduces the five observations provided for this analysis. The ratios are calculated from the supplied heights using Equation (9).

| Tree | $H$ (m) | $h$ (m) | $R = (H-h)/H$ | Reference |
|---|---|---|---|---|
| 1 | 7.80 | 4.50 | 0.42308 | $\sqrt{2}-1$ |
| 2 | 8.50 | 5.00 | 0.41176 | $\sqrt{2}-1$ |
| 3 | 7.20 | 4.20 | 0.41667 | $\sqrt{2}-1$ |
| 4 | 9.00 | 5.40 | 0.40000 | $\sqrt{2}-1$ |
| 5 | 7.50 | 4.40 | 0.41333 | $\sqrt{2}-1$ |

Table 1: Reported pilot measurements and crown ratios. The theoretical reference is $R_0 = \sqrt{2} - 1 \approx 0.414214$.

# 4 Statistical Analysis: TOST Equivalence Procedure

## 4.1 Hypotheses and equivalence margin

The analysis uses the Two One-Sided Tests (TOST) procedure for equivalence testing, following Schuirmann [1]. The stated equivalence margin is

$$\Delta = 0.03, \tag{11}$$

so the equivalence interval around the theoretical reference is

$$[R_0 - \Delta,\ R_0 + \Delta] \approx [0.384214,\ 0.444214]. \tag{12}$$

Let $\mu$ denote the population mean under the sampling model. The TOST hypotheses are

$$H_{01} : \mu \leq R_0 - \Delta, \tag{13}$$
$$H_{02} : \mu \geq R_0 + \Delta, \tag{14}$$

against the alternative that the mean lies strictly within the equivalence limits.

At significance level $\alpha = 0.05$, equivalence is concluded only when both one-sided null hypotheses are rejected. For the conventional paired interpretation of the TOST and the corresponding confidence-interval method, the 90% confidence interval for the mean must lie entirely within the equivalence bounds.

The equivalence margin is a substantive tolerance, not a quantity determined by the observed data. Its scientific justification should be specified independently of the observed results. If the margin was selected after inspecting the sample, it should be described as exploratory rather than prespecified.

## 4.2 Descriptive statistics

Using the five ratios in Table 1, the reported descriptive statistics are approximately

$$\begin{aligned} n &= 5, \\ \bar{R} &= 0.41297, \\ s &= 0.00845, \\ SE &= 0.00378. \end{aligned}$$

Under a one-sample Student $t$ model with $df = n - 1 = 4$, the reported 90% confidence interval is approximately

$$[0.40491,\ 0.42102]. \tag{15}$$

This interval lies within the equivalence bounds in Equation (12). The reported TOST p-value is approximately 0.000801, using the larger of the two one-sided p-values as the combined TOST p-value.

The numerical result should be interpreted conditional on the sampling design and statistical assumptions. In particular, the analysis assumes that observations are independent and that a one-sample $t$ procedure is appropriate for the population mean. With only five observations, distributional assumptions and outlier sensitivity cannot be assessed reliably.

## 4.3 Interpretation and limitations

The TOST result indicates that the sample mean is statistically equivalent to the theoretical reference within the stated margin under the assumptions of the test. It does not demonstrate

that individual trees have this ratio, that the ratio is biologically optimal, or that solar geometry causes the observed morphology.

The small sample size and single-location sampling make this a preliminary exploratory analysis. A stronger follow-up study should use a clearly defined sampling frame, more trees from multiple locations, repeatability checks, an independently justified equivalence margin, and an *a priori* sample-size or power analysis.

If several trees share a site or management history, the analysis should also consider whether observations are clustered rather than independent.

## 5 A Conjectural Numerical Construction Involving the Riemann Zeta Function

This section presents a numerical modeling proposal rather than a proven connection between tree geometry and the Riemann zeta function.

Let $\zeta(s)$ denote the Riemann zeta function. Write $\gamma_2 \approx 21.022039$ for the ordinate of the second positive nontrivial zero on the critical line, where zeros are conventionally expressed as $s = \frac{1}{2} + i\gamma$ and ordered by increasing positive ordinate. The first positive ordinate is approximately 14.134725, and the second is approximately 21.022039 [2].

Define the integer parameter

$$x_p = \lfloor \gamma_2 \rfloor = 21. \tag{16}$$

Within the proposed numerical model, the parameter $x_p$ is used in the expression

$$Z(x) = x^{-5} \ln(x) \sin\left(\frac{2\pi}{x}\right) e^{-x/x_p}, \qquad x > 1. \tag{17}$$

Equation (17) defines a model expression. The present paper does not derive this expression from properties of $\zeta(s)$, nor does it establish that it describes a physical field.

The parameter $x_p = 21$ is referred to within the Al-Hindara project as the *Yemeni Constant*. This is a project-specific name for the chosen integer parameter, not a recognized mathematical constant established by the results of this paper.

The model also considers the numerical quantity

$$K = 2\pi e^{-1/42} \approx 6.1353. \tag{18}$$

The expression in Equation (18) is mathematically well-defined, but its occurrence alone does not establish a connection to the asymptotic behavior of zeta zeros or to the Riemann critical line.

In particular, the proposed location $2x_p = 42$ for a peak of an accumulation function cannot be verified here because that function has not been explicitly defined and analyzed. Establishing such a claim would require a precise definition of the accumulation function, a domain, appropriate regularity conditions, and a derivation locating and classifying its extrema.

Accordingly, the recurrence of the integers 21 and 42 is presented as a conjectural feature of the proposed numerical construction. No theorem linking these integers to the geometric ratio $\sqrt{2} - 1$ or to the distribution of nontrivial zeta zeros is claimed.

## 6 Discussion

The results of this paper belong to three distinct categories.

First, the half-angle identity in Equation (8) is exact and follows from standard trigonometry.

Second, the five reported crown ratios produce a sample mean close to the geometric reference value. The equivalence analysis quantifies this closeness relative to the chosen margin, but it cannot establish a universal biological pattern from such a small sample.

Third, the numerical construction involving $x_p = 21$ and $K = 2\pi e^{-1/42}$ is conjectural. It is a proposal for future analysis, not a consequence of the geometric proof or the field measurements.

These distinctions are essential to the interpretation of the Al-Hindara framework. Conceptual unity can motivate mathematical models, but a quantitative relation requires operational definitions, reproducible measurements, and a derivation or empirical test appropriate to the claim.

## 7 Conclusion

This paper derives the exact identity

$$\tan\left(\frac{\pi}{8}\right) = \sqrt{2} - 1,$$

which is the reciprocal of the conventional Silver Ratio. Under the specified angle convention, the result gives an exact shadow-to-height ratio for an idealized right triangle.

The five reported mango-tree observations have a mean crown ratio of approximately 0.41297. The reported TOST analysis indicates equivalence to the theoretical reference within the stated margin of $\pm 0.03$, subject to the assumptions and limitations of the analysis. Because the sample is small and drawn from one location, the result is exploratory and should not be interpreted as evidence of a universal biological law or a causal relationship.

The paper also defines a conjectural numerical construction using the integer $x_p = 21$. Its proposed relationship to the Riemann zeta function remains unproved and requires explicit mathematical definitions and independent derivation.

Finally, the relation $Z + C + A = 1$ remains a conceptual normalization proposal until its variables and normalization rules are formally specified. Future work should prioritize reproducible field methods, larger and more diverse samples, and rigorous analysis of the proposed numerical model.

## References

[1] Schuirmann, D. J. (1987). A comparison of the two one-sided tests procedure and the power approach for assessing the equivalence of average bioavailability. *Journal of Pharmacokinetics and Biopharmaceutics*, 15(6), 657–680.

[2] Edwards, H. M. (1974). *Riemann's Zeta Function*. Academic Press.

[3] Hallé, F., Oldeman, R. A. A., and Tomlinson, P. B. (1978). *Tropical Trees and Forests: An Architectural Analysis*. Springer-Verlag.

## A Supplementary Web Applications

The following web applications are listed as supplementary project resources. Their availability and calculations should be independently checked before publication.

- **Zenodo Record (This Paper):**
  https://zenodo.org/records/23540796
- **DOI (This Paper):**
  https://doi.org/10.5281/zenodo.23540796
- **Frame 2 DOI (OSF):**
  https://doi.org/10.17605/OSF.IO/YC9MB

- **Visual Simulation:** https://jabri-com.vercel.app/tree/tree2.html
- **TOST Field Tool:** https://jabri-com.vercel.app/tree/tree3.html

## B Reproducibility Notes

For a reproducible statistical report, the analysis archive should include the unrounded measurements, the calculation of each ratio, the sample standard deviation, the standard error, the confidence interval, and the two one-sided test statistics and p-values.

The reported summary values in this manuscript are rounded for presentation. Calculations should be performed using unrounded ratios derived from the original measurements. The measurement record should also document the actual instrument and field procedure used.